

Assignment 8:

OBJECTIVES:

- **Understand BGP Fundamentals:** To gain a working knowledge of how the **Border Gateway Protocol (BGP)** operates as a path vector routing protocol.
- **Configure eBGP:** To successfully set up **External BGP (eBGP)** connections to facilitate communication between different Autonomous Systems (AS 100, AS 200, and AS 300).
- **Analyze Path Selection:** To evaluate how BGP selects the best routing paths using specific **BGP attributes** (such as AS_PATH, NEXT_HOP, LOCAL_PREF, and MED).
- **Implement Network Topology:** To build a functional network in Cisco Packet Tracer involving three routers, each representing a distinct administrative domain.
- **Verify Routing Information:** To use verification commands (like show ip bgp and show ip route) to ensure the routing table is correctly populated and loop-free

Router Configurations:

```
enable
configure terminal
interface gig0/0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

interface gig0/0/1
ip address 10.0.0.1 255.255.255.252
clock rate 64000
no shutdown
exit

ip route 192.168.2.0 255.255.255.0 10.0.0.2

router bgp 100
neighbor 10.0.0.2 remote-as 200
network 192.168.1.0 mask 255.255.255.0
end
```

```
enable
configure terminal
interface gig0/0/0
 ip address 10.0.0.2 255.255.255.252
 no shutdown
exit

interface gig0/0/1
 ip address 10.0.0.5 255.255.255.252
 clock rate 64000
 no shutdown
exit

ip route 192.168.1.0 255.255.255.0 10.0.0.1
ip route 192.168.2.0 255.255.255.0 10.0.0.6

router bgp 200
 neighbor 10.0.0.1 remote-as 100
 neighbor 10.0.0.6 remote-as 300
 network 10.0.0.0 mask 255.255.255.252
end
```

```
enable
configure terminal
interface gig0/0/1
 ip address 10.0.0.6 255.255.255.252
 no shutdown
exit

interface gig0/0/0
 ip address 192.168.2.1 255.255.255.0
 no shutdown
exit

ip route 192.168.1.0 255.255.255.0 10.0.0.5

router bgp 300
 neighbor 10.0.0.5 remote-as 200
 network 192.168.3.0 mask 255.255.255.0
end
```